

Logan Hysen

Doctor of Philosophy Candidate – Department of Fisheries and Wildlife
Center for Systems Integration and Sustainability (CSIS)
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Research Interests

My research interests are focused on using spatial methods to understand interactions at the **intersection of wildlife, humans, and the environment**. I ask questions related to promoting both conservation and human well-being simultaneously.

Education

Michigan State University	Expected May 2028
Dual Ph.D. Fisheries and Wildlife, Environmental Science and Policy	
GPA: 4.0/4.0 cumulative	
California State Polytechnic University, Humboldt (formerly Humboldt State University)	May 2023
M.Sc. Wildlife Ecology	
GPA: 4.0/4.0 cumulative	
University of Tennessee, Knoxville, TN	May 2020
B.S. Environmental and Soil Sciences, Environmental Science Concentration	
Watersheds minor	
GPA: 3.96/4.0 cumulative, <i>summa cum laude</i> , Top Upperclass Scholar	

Research Experience

Graduate Research Assistant, Michigan State University Center for Systems Integration and Sustainability, Department of Fisheries and Wildlife Advisor: Dr. Jianguo Liu	August 2023 – Present
- Studied the interaction between humans and Giant Panda (<i>Ailuropoda melanoleuca</i>) habitat in southeastern China - Developed surveys for households in rural China to understand human-landscape interactions and how they are mediated by livelihoods and environmental factors, ultimately influencing Giant Panda habitat conservation - Developed protocols and trainings for 40+ field team members - Managed, organized, and analyzed large survey data covering 37 nature reserves, 2,800 households, and over 600 unique columns of data per household - Organized collection of vegetation data across southeastern China with the goal of assessing Giant Panda habitat quality - Collected, prepared, and analyzed remote sensing data from diverse sources - Connected remote sensing and spatial data with human livelihood and land use data to understand how economic and environmental forces influence human land use decisions and ultimately forest cover and panda habitat	
Graduate Research Assistant, California State Polytechnic University, Humboldt Department of Wildlife (Arcata, CA) Advisor: Dr. Ho Yi Wan	August 2021 – May 2023
Performed spatial analyses (e.g., habitat suitability, connectivity modeling) for northern spotted owls using ensemble learning and connectivity software (e.g., Circuitscape, FRAGSTATS)	

- Investigated the presence-only ensemble habitat selection model development process and developed recommendations for best practices
- Published four peer-reviewed manuscripts related to spatial wildlife conservation and ecology
- Mentored undergraduate students in spatial analysis

Undergraduate Lab Assistant, University of TN, Knoxville

January 2019 – June 2019

Department of Ecology and Evolutionary Biology (Knoxville, TN)

- Collected data on larval and adult dung beetles for use in a research study, including monitoring gas exchange and body mass
- Assisted research team with tasks necessary to ensure a clean, functioning research environment

EPA Rain Works competition

Fall 2018

Project Title: *Marbled Ecologies*

- Used ArcGIS and Python programming to perform spatial analysis of habitat interconnectivity, stormwater discharge, and pollutant loading for a site on the campus of the University of Tennessee
- Coordinated the completion of tasks by members of an interdisciplinary team

Peer-Reviewed Publications (n citations = 113, *indicates co-first author, †indicates undergraduate mentee)

Google Scholar Link: <https://scholar.google.com/citations?user=1rHLFdoAAAAJ&hl=en>

1. House, Q., **Hysen, L.**, & Wan, H. Y. (2026). Simulating present and future connectivity of a threatened native owl under invasion by an exotic owl. *Biological Conservation*. <https://doi.org/10.1016/j.biocon.2026.112014>
2. Wan, H. Y. & **Hysen, L.** (2026). No Coding, No Problem: Revolutionizing Scientific Workflows through Workflow-orchestrating AI. *The Stacks*. <https://doi.org/10.60102/stacks-26005>
3. **Hysen, L.***, Nayeri, D.†, Pilgreen, D., Rastgoo, R., Sardari, P., Mohammadi, A., Kashfi, S. M., Dayer, A., & Kyle, G. (2025). Socio-psychological determinants of beekeeper conflict with bee-eaters. *Conservation Science and Practice*. <https://doi.org/10.1111/csp2.70219>
4. **Hysen, L.**, Wan, H. Y., Jantz, P., Gagnon, J., & Cushman, S. (2025). Variability in habitat selection between herds for a widespread ungulate. *Ecological Modelling*. <https://doi.org/10.1016/j.ecolmodel.2024.110991>
5. Nayeri, D., Cushman, S., Ganey, J., **Hysen, L.**, Gunther, M. S., Willey, D., & Wan, H. Y. (2024). Multiscale habitat suitability modeling for a threatened raptor offers insight into ecological model transferability. *Ecological Modelling*. <https://doi.org/10.1016/j.ecolmodel.2024.110845>
6. Wang, Z., Wang, Y., Cahill Jr., J., Li, X., Wan, H. Y., **Hysen, L.**, Wang, D., & Zhong, Z. (2023). Standing litter modifies top-down effects of large herbivores on grassland plant community. *Ecosystems*. <https://doi.org/10.1007/s10021-023-00864-y>
7. **Hysen, L.**, Nayeri, D., Fogarty, F., Kelly, E., Cushman, S., & Wan, H. Y. (2023). Northern spotted owl nesting habitat under high potential wildfire threats along the California Coastal Redwood Forest. *Science of the Total Environment*. <https://doi.org/10.1016/j.scitotenv.2023.163414>
8. **Hysen, L.**, Nayeri, D., Cushman, S. A., & Wan, H. Y. (2022). Background sampling for multiscale ensemble habitat selection modeling: how many points should be used? *Ecological Informatics*. <https://doi.org/10.1016/j.ecoinf.2022.101914>
9. Nayeri, D., Mohammadi, A., **Hysen, L.**, Hipólito, D., Huber, D., & Wan, H. Y. (2022). Identifying human-caused mortality hotspots to inform human-wildlife conflict mitigation, implications for conservation. *Global Ecology and Conservation*. [10.1016/j.gecco.2022.e02241](https://doi.org/10.1016/j.gecco.2022.e02241)

Submitted Manuscripts

1. Rodriguez Gutierrez Estrada†, J., **Hysen, L.**, Nayeri, D., & Wan, H. Y. In review. Back in the wild: where reintroduced California condors spend their time. *Ibis*.
2. Zhang, Y., Wu, H., Wu, Y., **Hysen, L.**, Liu, Y., Zhang, H., Yang, J., Fu, L., Xiao, M., Lu, L., Zhu, D., & Zhang, Z. In review. Human activities in different types of protected areas shape the patterns of args in wildlife gut microbiomes.
3. **Hysen, L.**† & Van Zeghbroeck, J.* In review. International food trade influences sustainable development progress. *Geography and Sustainability*.

Oral Presentations (* denotes invited talks)

1. **Hysen, L.**, Vina, A., Jia, N., Li, S., Nichols, S., Yang, H., Zhang, J., Ouyang, Z., Liu, J. (2026) Metacoupling in Panda Landscapes: Aligning Forest Conservation and Human Wellbeing. North American Regional Association of the International Association for Landscape Ecology (IALE-NA), Athens, GA
2. **Hysen, L.**, Cao, H., Huang, J., Jia, N., Manning, N., Remer, M., Van Zeghbroeck, J., Waloven, Kapsar, K., Wan, H.Y., & Liu, J. (2025) The First Decade of Telecoupling Research. North American Regional Association of the International Association for Landscape Ecology (IALE-NA), Raleigh, NC
3. **Hysen, L.**, Akande, H., Nayeri, D., & Collier, C. (2024) How are Species Distribution Models Informing Worldwide Management and Conservation? North American Regional Association of the International Association for Landscape Ecology (IALE-NA), Oklahoma City, OK
4. **Hysen, L.** & Liu, J. (2024) Global Protected Areas and the Biodiversity They Protect: How are they Influenced by Distant Relationships? Fisheries and Wildlife Graduate Research Symposium, Michigan State University, Lansing, MI
5. **Hysen, L.**, Cushman, S., Nayeri, D., & Wan, H. Y. (2023) Sampling the Background: Which strategy should you use? North American Regional Association of the International Association for Landscape Ecology (IALE-NA), Riverside, CA
6. Nayeri, D., **Hysen, L.**, Willey, D., Ganey, J., Cushman, S., & Wan, H. Y. (2023) Characterizing bimodal habitat selection of the Mexican spotted owl with multiscale ensemble modeling. North American Regional Association of the International Association for Landscape Ecology (IALE-NA), Riverside, CA
7. **Hysen, L.**, Nayeri, D., Ganey, J., Cushman, S., & Wan, H. Y. (2022) A tale of two owls: using multi-scale optimization models to identify nuances in habitat selection between two owl subspecies. North American Regional Association of the International Association for Landscape Ecology (IALE-NA), virtual
8. Nayeri, D., **Hysen, L.**, Willey, D., Ganey, J., Cushman, S., & Wan, H. Y. (2022) One species, two worlds: identifying habitat covariates and optimal scales of Mexican spotted owl habitat selection between canyonland and forest habitats. North American Regional Association of the International Association for Landscape Ecology (IALE-NA), virtual
9. ***Hysen, L.** & Nayeri, D. (2022) When two worlds collide: characterizing and predicting the patterns of human-caused mortality of a large carnivore. Conservation Unlimited Tuesday Talks, Cal Poly Humboldt, Arcata, CA
10. **Hysen, L.**, Nayeri, D., Ganey, J., Cushman, S., & Wan, H. Y. (2022) Using multi-scale optimization models to identify differences in habitat selection between two owls. North American Congress for Conservation Biology (NACCB), Reno, NV
11. Nayeri, D., **Hysen, L.**, Willey, D., Ganey, J., Cushman, S., & Wan, H. Y. (2022) Unraveling optimal scales and habitat covariates of Mexican spotted owl habitat selection between two different landscapes. North American Congress for Conservation Biology (NACCB), Reno, NV

Other Professional Positions

Game Pen Technician

September 2021 – May 2022

Humboldt State University Game Pens (Arcata, CA)

Supervisor: Jaclyn Patmore

Field Technician

June 2018 – August 2021

Grow Environmental Solutions, LLC. (Tullahoma, TN)

Supervisor: Christopher Grow

- Performed Rare, Threatened, and Endangered species surveys, habitat assessments, mitigation project assessments, wetland delineations, hydrologic determinations, communicated with government agencies
- Responsible for compiling reports, technical documents, and associated GIS data
- Developed novel GIS data analysis pipelines for use in the field

Funded Grants (Funding to Date: \$1,089.37)

California North Coast Chapter of The Wildlife Society (Co-Principal Investigator)	April 2022
Amount: \$1,089.37	
Research Topic: To conduct a study on demographics of northern spotted owl prey species in partnership with the United States Forest Service	

Unfunded Proposals

Save the Redwoods League Research Grant (<i>Invited to Submit Full Proposal</i>)	September 2021
Amount: \$25,000, unfunded	
Research Topic: To assess fuel loading, habitat, and movement corridors in Northern California	
Yolo Basin Graduate Research Fellowship	June 2021
Amount: \$12,000, unfunded	
Research Topic: To develop wetland connectivity models for avian and amphibian species in the Yolo Basin	
National Science Foundation Graduate Research Fellowship Program	October 2021
Research Topic: To assess forest fuels and develop forest fuel models for northern California to model wildfire risk, unfunded	

Professional Service

Reimaging Graduate Education within CANR Student Representative, Michigan State University	2026-present
Responsible for representing students from the College of Agriculture and Natural Resources and providing input on graduate student-related policy changes during the college's graduate program overhaul.	
Co-Leader, " <i>Telecoupling and Metacoupling: New Frontiers for Global Sustainability</i> " Workshop, International Association of Landscape Ecology, North America	2024, 2025, 2026
Fisheries and Wildlife Graduate Student Organization Symposium Co-Chair, Michigan State University	2024-2025
Fisheries and Wildlife Graduate Committee Student Representative, Michigan State University	2023-present
Vice President, Wildlife Graduate Student Society, Cal Poly Humboldt	2022-2023
Reviewer for the following peer-reviewed journals: <i>Ecological Informatics, Ecological Modelling, Journal for Nature Conservation, Southwestern Naturalist, Stacks</i>	

Awards/Recognitions/Scholarships

NASA-MSU Professional Enhancement Award (\$800.00)	2025
Best Student Presentation, Fisheries and Wildlife Graduate Research Symposium, MSU (\$250.00)	2024
University Distinguished Fellowship, Michigan State University (\$106,000.00)	2023-2027
James R. Koplin Outstanding Graduate Student Award, Wildlife Department, Cal Poly Humboldt (\$---	2023
Wildlife Graduate Student Society Mini-Grant, Cal Poly Humboldt (\$400.00)	2023
Student Travel Award – International Association of Landscape Ecology, North America (\$800.00)	2023
Association of Environmental Professionals Bay Area Student Scholarship (\$1,000.00)	2022
Geoffrey C. Bain Jr. Memorial Scholarship, College of Natural Resources & Sciences (\$2,000.00)	2022
Student Travel Award – North American Congress for Conservation Biology (\$250.00)	2022
Student Travel Award – International Association of Landscape Ecology, North America (\$135.00)	2022
Graduate Assistant Tuition Waiver, Wildlife Department, Cal Poly Humboldt (\$7,200.00)	2022
Student Travel Award – Western Section of The Wildlife Society (\$215.00)	2022
Environmental and Soil Sciences Top Upper-class Scholar Award, University of TN, Knoxville (\$200.00)	2019-2020
3rd Place Poster, 7th Annual Watershed Symposium at the University of TN, Knoxville (\$50.00)	2019

Certifications

Certificate in College Teaching, Michigan State University	In progress
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Teaching Experience

Instructor, Frontiers in Metacoupling Summer School	August 2026
Beijing Normal University, Beijing, China	
Graduate Assistant, Ornithology (WLDF 365)	January 2022 – May 2022
Department of Wildlife (Arcata, CA)	
Primary Instructor: Dr. Frank Fogarty III	
Duties included assisting during lab instruction and field trips, helping individual and groups of students understand lab concepts, and grading assignments and exams	
Graduate Assistant, Wildlife Field Techniques (WLDF 311)	August 2022 – December 2022
Department of Wildlife (Arcata, CA)	
Primary Instructor: Dr. Nicholas Kerhoulas	
Duties include assisting during lab instruction and field trips, grading assignments and exams, assist during preparation of lab and field trip material	

Professional Membership

The Wildlife Society	2022 – Present
Northern American Regional Association of the International Association for Landscape Ecology	2022 – Present
Society for Conservation Biology	2022 – Present
California North Coast Chapter of the Wildlife Society	2022 – 2023

Skills

Modeling: Multi-scale optimized habitat selection, machine learning (random forest & artificial neural networks), ensemble learning, CDMetaPOP, MaxEnt, statistical models (e.g., GLMM, GLM, GAM), UNICOR, home range estimation, JAGS, PATHWALKER, Circuitscape

GIS: R, Python, Google Earth Engine, FRAGSTATS, ArcGIS

Programming: R, Python, JavaScript, Julia (beginner), html (proficient)

Field: Amphibian and small mammal handling, questionnaire design, mist netting, camera trapping, habitat surveys, belt transects, flora & fauna of the U.S. identification, Survey123, wetland delineation

Languages: English (Native), Spanish (Intermediate), Farsi (Intermediate), Albanian (Intermediate), Mandarin Chinese (Novice)

Other Experience

Hub Coordinator for Sunrise Movement, Knoxville, TN

January 2019 – July 2020

- Founded an organization focused on climate change outreach to various groups
- Organized events related to environmental protection, communication, inclusivity, and non-violence